WORKED SOLUTION

STEP I Specimen paper, Question 12

Mechanics • Collisions with Friction

STEP I Specimen paper, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q12

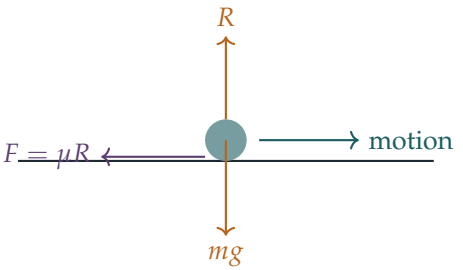
The key idea

Friction does all the work here. A disc sliding on level ice feels a constant retarding force, so it has constant deceleration, and the motion between events is just "constant acceleration" in disguise. We work backwards from the end: B's stopping distance fixes its speed after the impact, the collision laws turn that into A's speed just before the impact, and one more constant-acceleration step takes us back to the start at O.

Solution

The model

Treat each disc as a particle of mass m on a horizontal rough surface with coefficient of friction μ. While a disc slides, the forces on it are its weight mg downwards, the normal reaction R upwards and limiting friction F opposing the motion.



Resolving vertically, there is no vertical acceleration, so $R = mg$. The disc is sliding, so friction is limiting: $F = \mu R = \mu mg$. Resolving horizontally, $m\ddot{x} = -\mu mg$, so every sliding disc decelerates at the constant rate

$$a = \mu g,$$

whatever its mass. We also assume the collision is instantaneous, so friction has no time

to act during the impact itself; only the collision forces between the discs matter then, and momentum along OC is conserved.

Speed of B after the collision

B starts from the point at distance d from O and stops at C, at distance $2d$ from O, so it slides a distance d. Let its speed just after the collision be w. Using $v^2 = u^2 + 2as$ with final speed 0, acceleration $-\mu g$ and displacement d,

$$0 = w^2 - 2\mu g d \quad \Longrightarrow \quad w = \sqrt{2\mu g d},$$

which is the required result.

Speed of A just before the collision

Let A hit B with speed u, and let A's velocity just afterwards be v_A (both measured in the direction $O \to C$). Conservation of momentum, with equal masses m, gives

$$mu = mv_A + mw \quad \Longrightarrow \quad u = v_A + w,$$

and Newton's experimental law gives

$$w - v_A = eu.$$

Adding these eliminates v_A: $2w = (1+e)u$, so

$$u = \frac{2w}{1+e} = \frac{2\sqrt{2\mu g d}}{1+e}.$$

(Subtracting instead gives $v_A = \frac{1}{2}(1-e)u \geq 0$, so A carries on in the same direction more slowly.)

Ask yourself Could A catch B again and spoil the calculation?

No. After the impact both discs decelerate at the same rate μg, so the gap between their speeds, $w - v_A = eu$, stays constant while both are moving. B is never slower than A, so it stays ahead; and A, starting slower, stops first. So there is exactly one collision, and B's journey from d to $2d$ is undisturbed. (When $e = 0$ the discs move together until they stop, still with no further impact.)

The starting speed

A slides a distance d from O to B, decelerating at μg, so by $v^2 = u^2 + 2as$ again,

$$u^2 = U^2 - 2\mu g d \quad \Longrightarrow \quad U^2 = u^2 + 2\mu g d.$$

Substituting for u,

$$\begin{aligned} U^2 &= \frac{4 \cdot 2\mu g d}{(1+e)^2} + 2\mu g d = \frac{2\mu g d}{(1+e)^2}(4 + (1+e)^2) \\ &= \frac{2\mu g d}{(1+e)^2}(5 + 2e + e^2). \end{aligned}$$

Taking the positive square root, as $U > 0$ and $1 + e > 0$,

$$U = \frac{\sqrt{2\mu g d}}{1+e}\,(5 + 2e + e^2)^{\frac{1}{2}},$$

as required.

Checks

- Perfectly elastic, $e = 1$: identical discs swap velocities, so A stops dead and B leaves with A's arrival speed $u = w$. The formula gives $U = \sqrt{2}\,\sqrt{2\mu g d} = \sqrt{2\mu g (2d)}$, exactly the speed needed for a single disc to slide the whole $2d$ to C. That makes sense: the momentum is passed straight on.
- Perfectly inelastic, $e = 0$: the formula gives $U = \sqrt{5}\,\sqrt{2\mu g d}$, larger than for $e = 1$, as it should be, since kinetic energy is lost in the impact. More generally $(5 + 2e + e^2)/(1 + e)^2 = 1 + 4/(1 + e)^2$ decreases as e increases.
- Dimensions: μ and e are pure numbers and $\sqrt{gd}$ is a speed.
- Numerically, with $\mu = 0.1$, $g = 9.8\ \mathrm{m\,s^{-2}}$, $d = 1$ m and $e = 0.5$, the formula gives $U = \frac{7}{3} \approx 2.33\ \mathrm{m\,s^{-1}}$. Stepping the motion forward in small time steps from this U (for many random choices of μ, d and e) brings B to rest within 0.001% of $2d$ from O.

Common slips

- Applying energy conservation across the collision. Unless $e = 1$, kinetic energy is lost in the impact; use momentum and the restitution law there, and the work–energy (or $v^2 = u^2 + 2as$) step only between events.
- Forgetting that A also slides a distance d before the impact, and writing $U = u$.
- Getting the restitution law the wrong way round: it is the speed of separation, $w - v_A$, that equals e times the speed of approach, u.

Beyond STEP

The tidy feature here is that friction gives every disc the same deceleration regardless of its mass, so in a frame decelerating at μg the ice is effectively frictionless while both discs slide, and the collision looks like a textbook one. Choosing a convenient frame is a central technique in undergraduate mechanics, and the friction law $F = \mu R$ itself is an empirical model whose limits are worth knowing. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both collisions and friction with care.

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